

RC-ZOQO: CERTIFYING RESOLUTION LIMITS IN QUANTIZED ZERO-TH-ORDER OPTIMIZATION

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ABSTRACT

Quantized zeroth-order (ZO) optimizers keep optimizing on a discrete grid, but once the nominal probe or update falls below one grid step the run enters a *one-grid regime*: queries keep flowing while the grid cannot express the intended motion. We identify and characterize this terminal resolution wall: for quadratics, exact phase-averaged floor laws show the RMS stationarity floor is linear in the grid spacing and curvature-weighted, so each extra bit approximately halves it. RC-ZOQO then audits, from function values alone, whether a γ -sufficient decrease remains observable, with an exact finite-sample certificate: the false-stop probability is $(1 - p_{\gamma,m}(x))^K$, and K failures certify $p_{\gamma,m}(x) \leq 1 - \delta^{1/K}$ at confidence $1 - \delta$. On INT4/INT8 MNIST MLPs the audit stops a resolution-exhausted full model (89.3%/33.8% fewer post-floor/total queries, no accuracy loss) and continues a head-only block, recovering +1.14 points a schedule stop discards. Code is available at https://github.com/ldx825/RC-ZOQO_ICASSP-2027.

Index Terms— zeroth-order optimization, quantization, low-bit training, black-box optimization, resolution certificate

1. INTRODUCTION

Quantized zeroth-order (ZO) adaptation makes forward-only fine-tuning possible at deployment precision: ZOQO keeps parameters, perturbations, and updates quantized [1]; QuZO uses stochastic rounding for low-bit large-model fine-tuning [2]; QZO perturbs quantization scales [3]; CAQ-ZO aligns the query geometry with the companding quantizer [4]; and forward-only fine-tuning is now routine [5], building on two-point ZO estimation [6, 7, 8, 9, 10], ZO-signSGD [11], sparse-gradient ZO [12], and evolution strategies [13]. These methods answer how to keep optimizing on a discrete grid; we ask *when the grid itself has become the bottleneck*. Once the nominal probe or update falls below one grid step, the loop can only round the intended motion away or force a one-grid update, while every iteration still costs queries. Forced one-grid updates can oscillate near a grid-misaligned optimum even under an exact sign oracle, and endpoint clipping can reverse a comparison; in our INT4 MLP run, post-entry forced updates spend 299 extra queries and drift 0.26 points.

A fixed schedule is a poor proxy: crossing the threshold is neither necessary nor sufficient for resolution exhaustion—it discards recoverable progress when the floor is not yet reached and keeps spending queries when it is. We exhibit both failure modes in one protocol: an exhausted full model, and a head-only block that crosses

the same threshold while one-grid decreases remain valuable (Section 5.2 recovers +1.14 points by auditing instead of stopping on schedule). One-grid entry should trigger a state check, not a stop.

We first price the grid, then turn the terminal regime into a black-box statistical decision. For quadratics, exact floor laws (Theorem 1) give the representable stationarity floor in closed form, $F_\Delta = \Delta \|H\|_F / \sqrt{12}$, with each extra bit approximately halving it. RC-ZOQO then *audits* the state (Definition 2): it tests up to K one-grid Rademacher directions at a fixed state and stops when none satisfies (3). No gradient, Hessian, or Lipschitz constant is needed—an exact chord decomposition (Lemma 1) uses only queried values—and the guarantee is a finite-sample statement about the audit distribution: if $p_{\gamma,m}(x)$ is the probability that a sampled direction is γ -useful, the false-stop probability is exactly $(1 - p_{\gamma,m}(x))^K$, and K failures certify $p_{\gamma,m}(x) \leq 1 - \delta^{1/K}$ at confidence $1 - \delta$ (Theorem 2); accepted steps form a finite sequence (Proposition 1).

Our contributions are threefold. (i) **Resolution wall characterization**: exact curvature-weighted floor laws, the bit law, spectral bounds for general curvature, and a range–resolution relation (Appendix). (ii) **A state-based black-box certificate**: the audit, its exact chord decomposition, the finite-sample certificate, the finite accepted-step bound, and resolution invariance. (iii) **Stop/continue validation**: on INT4/INT8 MNIST MLPs the audit stops an exhausted full model (89.3%/33.8% fewer post-floor/total like-for-like queries, no accuracy loss) and continues a head-only block, recovering +1.14 points a schedule stop discards.

RC-ZOQO is a diagnostic certificate for existing quantized ZO loops, not a replacement optimizer (cf. [14]); it does not remove the floor, it detects when a fixed-resolution run has reached it. Query budgets matter when every evaluation is paid for [15, 16]; sufficient-decrease polling is classical in direct search [17] and quantization error is classical [18]; ours is the quantized one-grid terminal regime.

2. BACKGROUND AND PROBLEM SETUP

2.1. Quantized ZO updates

Following the fully quantized setting of ZOQO [1], a uniform b -bit quantizer covers a range of width D with grid spacing $\Delta_b = D/(2^b - 1)$. Nominal probe/update scales are quantized to multiples of Δ_b ; ZOQO enforces one grid step once a nominal scale drops below the spacing, giving two queries and the clamped sign update $x_p^\pm = \text{clamp}(x \pm ar)$, $c = \text{sign}[f(x_p^+) - f(x_p^-)]$, $x_{\text{next}} \leftarrow \text{clamp}(x - \alpha cr)$, with quantized probe/update magnitudes a, α . Low-bit formats and quantized gradients are standard [19, 20, 21, 22, 23, 24, 25, 26, 27] (survey: [28]).

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Definition 1 (One-grid regime). *A quantized ZO run is in the one-grid regime when its nominal probe or update would fall below one grid spacing and the realized nonzero scale is one grid step ($a = \Delta_b$ and/or $\alpha = \Delta_b$). Crossing this threshold triggers auditing; it is not by itself a stopping criterion.*

2.2. Notation and assumptions

Let $x \in \mathbb{R}^d$ and let $\mathcal{G}_\Delta$ denote the scalar lattice of spacing Δ applied coordinate-wise, with x^* a continuous minimizer.

Assumption 1 (Random grid phase). *For phase-averaged floor calculations, the phase of x^* relative to each scalar cell is i.i.d. uniform in $[-\Delta/2, \Delta/2]$.*

Assumption 2 (Local smoothness and admissible queries). *For the audit analysis, f is L -smooth on every queried segment, and audit directions are restricted to coordinates with sufficient margin so that $x \pm ar$ and the one-grid update do not clip, isolating resolution effects from clipping; RC-ZOQO uses an active set of non-clipping coordinates.*

3. RESOLUTION LIMITS OF QUANTIZED ZO

3.1. Phase-averaged floor

For a quadratic objective $f(x) = \frac{1}{2}(x - x^*)^\top H(x - x^*)$ with $H \succeq 0$, the phase-averaged nearest-grid error $e = \bar{x} - x^*$ satisfies $\mathbb{E}[e] = 0$ and $\mathbb{E}[ee^\top] = \frac{\Delta^2}{12} I$ (Lemma 2, Appendix).

Theorem 1 (Curvature-weighted nearest-grid law). *Under Assumption 1,*

$$\mathbb{E}[f(\bar{x}) - f^*] = \frac{\Delta^2}{24} \text{tr}(H), \quad (1)$$

$$\mathbb{E}\|\nabla f(\bar{x})\|_2^2 = \frac{\Delta^2}{12} \text{tr}(H^2). \quad (2)$$

Hence the RMS nearest-grid stationarity scale is $F_\Delta = \frac{\Delta}{\sqrt{12}}\|H\|_F$. For diagonal H , nearest scalar quantization minimizes f over the grid, making (1)–(2) exact grid-constrained floors.

For non-axis-aligned curvature, Corollary 2 (Appendix) bounds the discrete optimum.

Combining Theorem 1 with $\Delta_b = D/(2^b - 1)$ yields the bit law $F_b = D\|H\|_F/(\sqrt{12}(2^b - 1))$, i.e., $b \geq \lceil \log_2(1 + D\|H\|_F/(\sqrt{12}\epsilon)) \rceil$ for target RMS stationarity scale ϵ . It is exact for axis-aligned quadratics, and $F_{b+1}/F_b \rightarrow 1/2$: each extra bit approximately halves the stationarity scale and quarters the objective scale.

Two examples show why the floor matters: for $f(x) = \frac{1}{2}(x - \Delta/2)^2$, an exact-sign one-grid update from $x_0 = 0$ alternates $0 \leftrightarrow \Delta$, so the best cell is reached yet forced updates continue; and for $f = x_1 + 0.75x_2$ on range $[0, 3a]$ at $x = (0, 1.5a)$, $r = (1, -1)$, the true derivative $0.25 > 0$ but the clamped comparison gives $-0.5a < 0$. RC-ZOQO avoids clipped chords and tests whether one-grid motion remains resolvable.

4. RESOLUTION-CERTIFIED ZOQO

4.1. Resolution audit

RC-ZOQO activates when the base quantized ZO optimizer reaches the one-grid regime; an audit direction then samples $m \leq q$ co-

ordinates uniformly without replacement from the admissible (non-clipping) set $\mathcal{A} \subseteq [d]$ ($q := |\mathcal{A}|$), with independent Rademacher signs ($r \in \{-1, 0, 1\}^d$, $\|r\|_0 = m$), probed symmetrically at magnitude a .

Definition 2 (Resolution audit). *For forcing constant $\gamma \geq 0$, an audit direction is declared successful if*

$$\min_{\sigma \in \{-1, +1\}} f(x + \sigma ar) \leq f(x) - \gamma a^2 m. \quad (3)$$

A successful direction is executed immediately: the better of $x \pm ar$ becomes the next iterate, so a successful audit both certifies and performs a one-grid decrease, reusing the queried point. RC-ZOQO tests up to K independent directions with (3); if all K fail, the run stops—a certificate on the sampled directions, not a proof that no progress exists. γ sets the minimum one-grid decrease deemed worth further queries, not whether any descent exists.

The forcing constant sets the gate’s strictness: $\gamma = 0$ certifies any strict decrease (on a noisy floor the audit never stops), larger γ stops earlier; the experiments report $\gamma \in \{0, 0.05, 0.25\}$. A self-calibrated γ^* from the one-grid two-point history $[(0.40, 0.62] \text{ INT4}, [1.36, 1.85] \text{ INT8})$ is a conservative heuristic, not an oracle for remaining progress.

The audit minimum over two one-grid moves has a closed form in the queried function values.

Lemma 1 (Exact chord decomposition). *For any direction r with $\|r\|_0 = m$, define the one-grid slope and curvature statistics $\hat{s}_r(x) = \frac{f(x+ar) - f(x-ar)}{2a}$ and $\hat{c}_r(x) = \frac{f(x+ar) + f(x-ar) - 2f(x)}{2a^2}$. Then*

$$f(x \pm ar) - f(x) = a^2 \hat{c}_r(x) \pm a \hat{s}_r(x),$$

$$\min_{\sigma \in \{-1, +1\}} f(x + \sigma ar) - f(x) = a^2 \hat{c}_r(x) - a |\hat{s}_r(x)|, \quad (4)$$

so (3) is equivalent to $|\hat{s}_r(x)| - a \hat{c}_r(x) \geq \gamma am$.

This test is the entire audit: $\hat{s}_r$ is the observable chord slope and $\hat{c}_r$ the curvature penalty paid on it, both computed from the two queried endpoints and the cached $f(x)$. No gradient, Hessian, global smoothness constant, or ambient dimension d enters the decision—only the m probed coordinates—which is why the audit remains executable at neural scale.

4.2. A finite-sample resolution certificate

What does a run learn from K failed audits? Define the sampled useful-direction mass $p_{\gamma, m}(x) = \Pr_r[A_r(x) = 1]$, where $A_r(x) = \mathbf{1}\{\min_\sigma f(x + \sigma ar) \leq f(x) - \gamma a^2 m\}$ and r is drawn from the audit distribution—a quantity defined purely by what the audit observes. The next result gives an exact finite-sample interpretation of K failed audits.

Theorem 2 (Finite-sample resolution certificate). *Fix x , $\gamma \geq 0$, m , and the audit distribution; assume failed audits leave x unchanged and directions are independent. Then the probability of observing K consecutive failures is exactly $(1 - p_{\gamma, m}(x))^K$; consequently, if $p_{\gamma, m}(x) \geq \epsilon$ for some $\epsilon \in (0, 1)$, the probability that RC-ZOQO stops after K failed audits is at most $(1 - \epsilon)^K$. Conversely, after observing K failures and no success, for any $\delta \in (0, 1)$ the one-sided bound $p_{\gamma, m}(x) \leq 1 - \delta^{1/K}$ holds with confidence at least $1 - \delta$.*

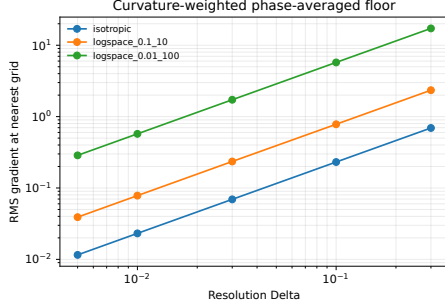


Fig. 1. RMS stationarity floor vs. grid spacing: unit log-log slope; curvature changes the coefficient.

Both parts are exact, with no smoothness or moment assumption; the theorem certifies scarcity of useful directions under the audit distribution, not the nonexistence of a descent direction. With $K=15$, $\delta=0.05$ this reads $p_{\gamma,m}(x) \leq 1 - 0.05^{1/15} \approx 18.1\%$; if $p_{\gamma,m}(x) \geq 0.2$, the false-stop probability is at most $0.8^{15} \approx 3.52\%$. Each audit costs at most $2K + 1$ evaluations per state: the cached $f(x)$ plus two probes per direction. A gradient-norm-only lower bound cannot generally remain dimension-free for sparse random audits, because a concentrated gradient may be missed by the sampled support; the Appendix records the classical condition and why it becomes vacuous at neural scale.

Proposition 1 (Finite audited-phase termination). *Suppose $\gamma > 0$ and $f \geq f_{\inf}$ on all queried states. Every accepted audit satisfies (3), so $N_{\text{acc}} \leq \lfloor (f(x_0) - f_{\inf})/(\gamma a^2 m) \rfloor$; after the last accepted step every audit fails, so RC-ZOQO stops within K audits with the certificate of Theorem 2. We deliberately do not claim convergence to a continuous stationary point: at fixed resolution the audited phase ends in a certified stop.*

The audit decision is a property of the state geometry, not the bit width: INT4 and INT8 runs can behave alike at matched normalized states.

Corollary 1 (Resolution invariance for quadratics). *Let $f(x) = \frac{1}{2}(x - x^*)^\top H(x - x^*)$, write $x = x^* + \Delta z$, and use a one-grid audit $a = \Delta$. Then (3) is equivalent to $|z^\top Hr| \geq \frac{1}{2}r^\top Hr + \gamma m$, which is independent of Δ ; if a ZO-sign step also moves exactly one grid step without clipping, then $z^+ = z - \text{sign}(z^\top Hr)r$, so the normalized dynamics are resolution-invariant.*

Inside the representable cell, no one-grid move strictly decreases a diagonal quadratic when $\gamma = 0$ (Appendix): the audit falls silent instead of forcing oscillatory moves.

5. EXPERIMENTS

For Theorem 1 we use $d \in \{16, 64, 256\}$, five resolutions, 5000 random phases, three Hessian spectra: relative errors of (1)/(2) are 1.09%/1.85%, and log-log RMS-floor slopes versus Δ are 0.99994–1.00041 (Fig. 1).

Theorem 2 is exact, so validation targets its inputs: the mass $p_{\gamma,m}(x)$ and the false-stop law $(1 - p_{\gamma,m}(x))^K$. Across 90 (q, k, m) configurations (q : coordinates, k : nonzero gradient entries, m : support size) at fixed quadratic states (10^5 draws each), empirical success frequencies reproduce the m/q support-hit scaling and the predicted phase transition, respecting the appendix’s gradient-based bound (no 3σ violation).

Table 1. MNIST MLP quantized-ZO adaptation (3 seeds): full model (top), head-only gate (bottom); “Post” = queries after one-grid entry.

Method	Bits	Acc. (%) $\uparrow$	Queries $\downarrow$	Post $\downarrow$
ZOQO	4	95.96	802	299
Clip-safe ZOQO	4	96.02	790	299
RC-ZOQO	4	96.22	523	32
ZOQO	8	96.81	802	299
RC-ZOQO	8	96.83	535	32

Head-only policy	Acc. (%) $\uparrow$	Queries $\downarrow$	Accepted
Stop at entry	93.85	118	0
Audit $\gamma = \gamma^*$, $m = 16$	93.85	149	0.0
Audit $\gamma = 0.25$, $m = 16$	94.26	178	2.7
Audit $\gamma = 0.05$, $m = 16$	94.99	256	21.7
Forced updates (no audit)	94.93	398	–
Audit $\gamma = 0$, $m = 64$	96.08	663	103.7

5.1. Stopping at a resolution floor

We use a 784–256–128–10 MLP on MNIST: the full-precision checkpoint reaches 96.72% and direct INT4 quantization 96.32% (INT8 essentially lossless). The black-box objective is cross-entropy on a fixed 4096-example subset (grid spacings $\Delta_4 = 0.0439$, $\Delta_8 = 0.0026$); all runs use 400 iterations, three seeds, and one-grid entry at iteration 251, where RC-ZOQO audits with $\gamma = 0.05$, $m = 1024$, and $K = 15$.

Pre-entry RC-ZOQO matches the clipping-safe baseline; fixed ZOQO then keeps forcing one-grid moves, drifting from 96.22% to 95.96% over 299 queries. RC-ZOQO rejects the first floor audit in all three seeds and stops: 32 post-floor queries instead of 299 (89.3% fewer) and 33.8% fewer total queries than the trajectory-matched clipping-safe baseline (523 vs. 790), avoiding the 0.26-point drift. All three seeds observe $K = 15$ consecutive failures, certifying at 95% confidence that fewer than 18.1% of sampled directions remain γ -useful. At $\gamma = 0$ it accepts one strict decrease, stalls, and ends at 95.22%: pure descent at a noisy floor is not accuracy-preserving. At INT8 the audit likewise spends 32 post-floor queries instead of 299 at comparable accuracy (96.83% vs. 96.81%), supporting the resolution-invariant interpretation.

5.2. Continuing despite schedule entry

To exercise the accepting side we repeat the protocol on the 1,290-parameter classifier head of the same MLP (one-grid entry at iteration 60, all audits at $K = 15$): INT4 quantization leaves the block at 96.68%, and the shared pre-floor trajectory ends at 93.85% when the one-grid regime begins, so the floor is not yet reached. The same $\gamma = 0.05$ rule that stopped the full model here continues: 21.7 accepted steps reach 94.99% at 256 queries, +1.14 points over the schedule stop (93.85% at 118). Table 1 traces the frontier: forced updates reach 94.93% at 398 queries, $\gamma = 0$ steps to 96.08% at 663, while $\gamma = 0.25$ stops after 2.7 accepted steps and γ^* stops at entry; at INT8 the gate stops after one accepted step (96.75% vs. 96.72%). “Accepted” counts executed steps; the 96.08% of $\gamma = 0$ is a looser policy than $\gamma = 0.05$. The audit is a state-based gate whose strictness sets the continue/stop frontier—not a schedule rule—and accepted steps pay their own cost. A $13\times$ larger LoRA-style block (16,640 parameters, rank 16 [29]) reproduces the stopping side ($\gamma \geq 0.05$ stops at entry; $\gamma = 0$ never stops). The two cases are the two sides of the same decision problem: one-grid entry triggers auditing, whereas the current state de-

cides between stop and continue.

Scope and limitations. RC-ZOQO diagnoses when a fixed-resolution run has reached the floor of Theorem 1; a failed audit certifies the absence of sampled γ -sufficient one-grid directions, not local optimality; the certificate needs no smoothness constant.

6. CONCLUSION

We identified a terminal resolution wall in quantized ZO: once the grid cannot express the intended motion, fixed-bit runs keep paying queries for updates that cannot carry signal. The wall has a computable price (curvature-weighted floor laws, approximately halved by each extra bit) and a testable state: the resolution audit turns the one-grid regime into a decision with an exact finite-sample certificate and a finite accepted-descent bound, stopping an exhausted full model (89.3%/33.8% fewer post-floor/total like-for-like queries, no accuracy loss) and continuing a head-only block, recovering +1.14 points a schedule stop discards.

7. APPENDIX: PROOFS

7.1. Proofs of Lemma 2 and Theorem 1

Lemma 2 (Random-phase moments). *Let $\bar{x}$ be the nearest-grid quantization of x^* and $e = \bar{x} - x^*$. Under Assumption 1, $\mathbb{E}[e] = 0$ and $\mathbb{E}[ee^\top] = \frac{\Delta^2}{12}I$.*

Proof of Lemma 2. For each coordinate, $e_j = \bar{x}_j - x_j^*$ is uniform on $[-\Delta/2, \Delta/2]$ by Assumption 1, so $\mathbb{E}[e_j] = 0$, $\mathbb{E}[e_j^2] = \Delta^2/12$, and $\mathbb{E}[e_j e_k] = 0$ for $j \neq k$; hence $\mathbb{E}[e] = 0$ and $\mathbb{E}[ee^\top] = (\Delta^2/12)I$.

Proof of Theorem 1. Since $e^\top H e = \text{tr}(H e e^\top)$, $f(\bar{x}) - f^* = \frac{1}{2} \text{tr}(H e e^\top)$; substituting Lemma 2 in expectation,

$\mathbb{E}[f(\bar{x}) - f^*] = \frac{1}{2} \text{tr}(H \mathbb{E}[ee^\top]) = \frac{1}{2} \text{tr}(H \frac{\Delta^2}{12}I) = \frac{\Delta^2}{24} \text{tr}(H)$, which is (1). Similarly $\nabla f(\bar{x}) = H e$ gives $\mathbb{E}\|\nabla f(\bar{x})\|_2^2 = \text{tr}(H^2 \mathbb{E}[ee^\top]) = \frac{\Delta^2}{12} \text{tr}(H^2)$, i.e., (2), and $F_\Delta = (\Delta/\sqrt{12})\|H\|_F$; for diagonal H each coordinate minimizes independently at the nearest grid point of x_j^* , making (1)–(2) exact.

7.2. Proof of Corollary 2 and auxiliary facts

Corollary 2 (General quadratic grid-floor bounds). *Under Assumption 1, for $H \succ 0$ with lattice minimizer x_Δ^* , writing $\lambda_- \leq \lambda_+$ for the extreme eigenvalues of H : $\mathbb{E}[f(x_\Delta^*) - f^*] \in \frac{d\Delta^2}{24}[\lambda_-, \lambda_+]$ and $\mathbb{E}\|\nabla f(x_\Delta^*)\|_2^2 \in \frac{d\Delta^2}{12}[\lambda_-^2, \lambda_+^2]$.*

Proof of Corollary 2. Let $\bar{x}$ be the coordinate-wise nearest lattice point of x^* , so $\delta_\Delta = \|\bar{x} - x^*\|$ and $\mathbb{E}[\delta_\Delta^2] = d\Delta^2/12$ (Lemma 2). *Objective:* since x_Δ^* minimizes f on the lattice, $f(x_\Delta^*) - f^* \leq \frac{1}{2}\lambda_+\delta_\Delta^2$; conversely, strong convexity and the lattice membership of x_Δ^* give $f(x_\Delta^*) - f^* \geq \frac{1}{2}\lambda_-\delta_\Delta^2$, giving the interval. *Gradient:* $\|\nabla f(x_\Delta^*)\|_2^2 = (x_\Delta^* - x^*)^\top H^2 (x_\Delta^* - x^*) \geq \lambda_-^2 \delta_\Delta^2$; and $H^2 \preceq \lambda_+ H$ (eigenvalues $\lambda_i^2 \leq \lambda_+ \lambda_i$) bounds the same quantity by $2\lambda_+[f(x_\Delta^*) - f^*] \leq \lambda_+^2 \delta_\Delta^2$; substituting $\mathbb{E}[\delta_\Delta^2]$ completes the interval.

Auxiliary facts. (a) A uniform b -bit quantizer over width D has step $\rho = D/(2^b - 1)$ and span $(2^b - 1)\rho$, so covering span W with margin h per side needs $W + 2h \leq (2^b - 1)\rho$. (b) For diagonal $H \succeq 0$, $|z^\top H r| \leq \sum_j |z_j| |h_j| |r_j| \leq \frac{1}{2} r^\top H r$ when $|z_j| < \frac{1}{2}$ on the support, and (4) at $\gamma = 0$ then keeps the audit minimum above $f(x)$: the audit falls silent inside the cell.

7.3. Proofs of Lemma 1, Thm. 2, Prop. 1, and Cor. 1

Proof of Lemma 1. The definitions give $\frac{1}{2}[f(x+ar) - f(x-ar)] = a\hat{s}_r$ and $\frac{1}{2}[f(x+ar) + f(x-ar) - 2f(x)] = a^2\hat{c}_r$; adding them gives $f(x+ar) - f(x) = a\hat{s}_r + a^2\hat{c}_r$, subtracting gives $f(x-ar) - f(x) = -a\hat{s}_r + a^2\hat{c}_r$, and minimizing over the sign gives $a^2\hat{c}_r - a|\hat{s}_r|$, so comparing with $-\gamma a^2 m$ yields the test.

Proof of Theorem 2. Failed audits leave x unchanged, so with $p = p_{\gamma,m}(x)$ and independence, $\Pr[K \text{ failures}] = (1-p)^K$; if $p \geq \varepsilon$ this is at most $(1-\varepsilon)^K$, bounding false stops. Conversely, if $p > 1 - \delta^{1/K}$ then $\Pr[K \text{ failures}] < \delta$; hence $p \leq 1 - \delta^{1/K}$ holds at confidence $1 - \delta$ (exact one-sided Clopper–Pearson).

Proof of Proposition 1. Each accepted audit decreases f by at least $\gamma a^2 m > 0$; telescoping gives $f(x_0) - f(x_{N_{\text{acc}}}) \geq N_{\text{acc}} \gamma a^2 m$, and $f \geq f_{\text{inf}}$ caps $N_{\text{acc}} \leq \lfloor (f(x_0) - f_{\text{inf}})/(\gamma a^2 m) \rfloor$. Each audit consumes one unit of the cap or fails; after K failures the run stops certified by Theorem 2.

Proof of Corollary 1. Substituting $x = x^* + \Delta z$, $a = \Delta$ gives $(z + \sigma r)^\top H(z + \sigma r) - z^\top H z = 2\sigma z^\top H r + r^\top H r$, so $f(x + \sigma \Delta r) - f(x) = \Delta^2(\sigma z^\top H r + \frac{1}{2} r^\top H r)$; the audit minimum is $\Delta^2(\frac{1}{2} r^\top H r - |z^\top H r|)$, so $|z^\top H r| \geq \frac{1}{2} r^\top H r + \gamma m$, free of Δ ; likewise the sign step $f(x + \Delta r) - f(x - \Delta r) = 2\Delta^2 z^\top H r$ gives $c = \text{sign}(z^\top H r)$ and $z^+ = z - \text{sign}(z^\top H r)r$ without clipping, also Δ -free.

7.4. Why the bound is vacuous at neural scale

Lemma 3 (Sparse-Rademacher sufficient decrease). *Under Assumption 2, (3) is guaranteed whenever $|g^\top r| \geq am(\gamma + L/2)$; for $Z = g^\top r$ under uniform m -subset support and i.i.d. Rademacher signs, $\mathbb{E}Z^2 = \frac{m}{q}\|g\|_2^2$, $\mathbb{E}Z^4 = \frac{m}{q}\|g\|_4^4 + \frac{3m(m-1)}{q(q-1)}(\|g\|_2^4 - \|g\|_4^4)$.*

Theorem 3 (Gradient-based sufficient condition). *Let $q \geq 2$, $g \neq 0$, $C = \gamma + L/2$, $\nu = \|g\|_4^4/\|g\|_2^4$, and*

$$\theta_m = \frac{a^2 m q C^2}{\|g\|_2^2}, \quad p_m = (1 - \theta_m)^2 \frac{m/q}{\nu + 3 \frac{m-1}{q-1} (1 - \nu)}. \quad (5)$$

If $\theta_m < 1$, a sampled direction succeeds with probability at least p_m ; for $m = q$, $\|g\|_2 \geq 2aqC$, $p_q \geq 3/16$, so $K = 15$ gives false-stop risk $(13/16)^{15} < 4.5\%$ (cf. [30, 8]).

Proof of Lemma 3. By L -smoothness, $f(x+ar) - f(x) \leq ag^\top r + \frac{1}{2}a^2m$, and for $-r$ likewise $f(x-ar) - f(x) \leq -ag^\top r + \frac{1}{2}a^2m$; hence $\min_\sigma f(x + \sigma ar) - f(x) \leq -a|g^\top r| + \frac{1}{2}a^2m \leq -\gamma a^2 m$ whenever $|g^\top r| \geq am(\gamma + \frac{L}{2})$. For the moments, condition on S : $Z = \sum_{j \in S} g_j \varepsilon_j$ has $\mathbb{E}[Z^2 | S] = \sum_{j \in S} g_j^2$ and $\mathbb{E}[Z^4 | S] = 3(\sum_{j \in S} g_j^2)^2 - 2 \sum_{j \in S} g_j^4$; averaging gives $\mathbb{E} \sum_{j \in S} g_j^2 = \frac{m}{q}\|g\|_2^2$ and $\mathbb{E}(\sum_{j \in S} g_j^2)^2 = \frac{m(m-1)}{q(q-1)}(\|g\|_2^4 - \|g\|_4^4) + \frac{m}{q}\|g\|_4^4$, i.e., the stated moments.

Proof of Theorem 3. Let $t = amC$; by Lemma 3 success is guaranteed for $|Z| \geq t$. For $t^2 < \mathbb{E}Z^2$, Cauchy–Schwarz gives $\Pr[|Z| \geq t] \geq (\mathbb{E}Z^2 - t^2)^2/\mathbb{E}Z^4$. With $\theta_m = t^2/\mathbb{E}Z^2 = a^2 m q C^2/\|g\|_2^2$ and $\nu = \|g\|_4^4/\|g\|_2^4$, $\mathbb{E}Z^2 - t^2 = \frac{m}{q}\|g\|_2^2(1 - \theta_m)$ and $\mathbb{E}Z^4 = \|g\|_4^4[\frac{m}{q}\nu + \frac{3m(m-1)}{q(q-1)}(1 - \nu)]$; substituting gives (5). For $m = q$, $\|g\|_2 \geq 2aqC$ gives $\theta_q \leq 1/4$ and $p_q \geq 3/16$, so K failures occur with probability at most $(13/16)^K < 4.5\%$ at $K = 15$.

Why the bound becomes vacuous. $\theta_m < 1$ requires $\|g\|_2 > aC\sqrt{mq} \propto \sqrt{q}$; at neural scale ($q \approx 2.4 \times 10^5$, $\Delta \approx 4 \times 10^{-2}$, $m \approx 10^3$) this needs $\|g\|_2 \gtrsim 6 \times 10^2 C$, so (5) degenerates; validation targets the certificate’s inputs instead.

8. REFERENCES

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